

UAV Swarm Networking: An MARL-Based Cross-Layer Transmission Framework

Yang Shen[✉], Graduate Student Member, IEEE, Bing Li[✉], Member, IEEE,
and Rongqing Zhang[✉], Senior Member, IEEE

Abstract—High-performance networking is essential for Unmanned Aerial Vehicle (UAV) swarms to accomplish complex, coordinated missions. A central challenge in UAV swarm networking is managing concurrent multi-hop transmissions, where traditional protocols often struggle due to routing path conflicts and co-channel interference. To address this, we propose a novel multi-agent reinforcement learning (MARL)-based cross-layer transmission framework that maximizes system throughput by jointly optimizing network-layer routing, link-layer resource allocation, and UAV trajectories. We decouple this complex joint optimization problem and solve it with a routing-prioritized iterative scheme. For the routing sub-problem, an MARL approach is designed for agents to collaboratively plan concurrent routing paths. The non-convex resource allocation and trajectory sub-problems are handled using successive convex approximation (SCA). Experimental results demonstrate that our proposed framework significantly outperforms existing benchmarks in system throughput, end-to-end delay, and packet delivery ratio.

Index Terms—Unmanned aerial vehicle (UAV), UAV swarm networking, cross-layer optimization, routing protocols, multi-agent reinforcement learning (MARL).

I. INTRODUCTION

IN RECENT years, unmanned aerial vehicles (UAVs) have been utilized in numerous applications, including logistics delivery [1], passenger services [2], and environmental monitoring [3]. In addition to these applications, UAVs also play a significant role in assisting communication. The flexibility of UAVs enables them to cover broader areas. Also, the high probability of line-of-sight (LOS) channels benefits air-to-ground communication. Therefore, UAVs can serve as aerial base stations [4], relays [5], and data collection devices [6]. No doubt that UAVs bring a new architecture to wireless networks.

Multiple UAVs can network together to accomplish sophisticated functions [7]. Compared to un-networked UAVs, a UAV swarm network can cover a larger area. In particular, UAV swarms can provide communication services in complex terrains or remote areas. When one UAV fails, other UAVs

can continue to carry out the mission, thereby enhancing the robustness of the network. Moreover, UAV swarms can exchange and aggregate information in real time, which improves the efficiency and accuracy of decision-making [8].

However, organizing large-scale UAV swarm networks presents technical challenges. To satisfy the long-distance communication demands between UAV nodes, the network needs to transmit data with multi-hop relaying. Due to the high mobility of UAVs, the network topology is unstable, which poses challenges to traditional routing protocols that are based on routing tables [9]. Moreover, tasks such as military operations, intelligent transportation, and augmented reality require lower transmission latency in UAV swarm networks [10]. Additionally, an efficient collaboration mechanism for UAVs is required to enhance network performance [11].

Current literature has proposed various solutions for these challenges, and performs well in handling one or a small number of routing requests. However, when facing a huge number of concurrent node-to-node communication demands, these methods encounter two critical issues. First, if several concurrent routing paths traverse the same node, they compete for communication resources, creating path conflicts that degrade the transmission rate. Secondly, when links on different paths are adjacent and transmit on the same resource blocks (RBs), co-channel interference may significantly increase. Owing to the bottleneck effect of multi-hop transmission, these two issues diminish overall system throughput. Addressing these concurrent communication challenges requires a holistic approach that jointly considers multiple paths and manages interference through cross-layer optimization.

Motivated by these issues, in this paper, we propose a cross-layer concurrent (CLACO) transmission framework that comprehensively takes into account network-layer routing paths, link-layer resource allocation, and UAV trajectories. The CLACO framework maximizes system throughput through a routing-prioritized iterative scheme. At its core, we employ a multi-agent reinforcement learning (MARL) approach to achieve collaborative routing. Experimental results show that CLACO significantly outperforms benchmarks regarding system throughput, end-to-end delay, and package delivery ratio. The main contributions are summarized as follows:

- To reduce routing conflicts and co-channel interference of concurrent multi-hop transmission in UAV swarms, we for the first time consider a cross-layer optimization of network-layer routing, link-layer resource allocation, and UAV trajectories to maximize system throughput.

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Yang Shen and Bing Li are with the School of Computer Science and Technology, Tongji University, Shanghai 200092, China (e-mail: stanleyshen@tongji.edu.cn; lizi@tongji.edu.cn).

Rongqing Zhang is with the Intelligent Transportation Thrust, The Hong Kong University of Science and Technology (Guangzhou), Guangzhou 511453, China (e-mail: rongqingz@hkust-gz.edu.cn).

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- To solve the joint optimization problem, we propose a cross-layer concurrent transmission framework that decouples it into three sub-problems and iteratively solves them with three corresponding modules. Given that routing paths play a crucial role in determining potential interference, the framework adopts a routing-prioritized iterative scheme to organize the three modules.
- In the routing module, considering that multi-hop routing involves sequential decision-making, we design an MARL-based collaborative routing (MACOR) algorithm implemented with proximal policy optimization (PPO). To encourage cooperation among routing paths, we allow data package agents to observe preceding paths and surrounding nodes, and define the reward function as the increment of system throughput.

The remainder of this paper is organized as follows. Section II summarizes the existing works in UAV swarm networking. Section III presents the system model and demonstrates the CLACO transmission framework. Section IV describes the proposed MARL-based collaborative routing algorithm in the routing module. In Section V, the UAV trajectory module and resource allocation module are introduced. Section VI reports the simulation results and compares CLACO with benchmarks. Finally, Section VII concludes the paper.

II. RELATED WORKS

The highly dynamic topology of UAV swarms poses significant challenges to networking, hindering the ability to support stable, low-latency, and efficient data transmission. This challenge is fundamentally a routing problem: how to find and maintain optimal data paths in a constantly changing environment. Therefore, research efforts have concentrated on achieving dynamic routing, low-latency transmission, and collaborative swarm routing. This section reviews notable works in these areas to contextualize our proposed framework.

A. Routing for Dynamic Topologies

To address the issue of unstable topology structure, several protocols have been developed. The routing protocol for autonomous UAV networks proposed in [12] introduced a multi-metric route selection mechanism combining hop count, route lifetime, estimated time to destination, and interfering links to choose stable, low-congestion paths. It featured preemptive route switching based on real-time link lifetime and congestion monitoring to avoid broken routes before packet loss occurred. Reference [13] developed an MARL routing algorithm for UAV swarm networks to address highly dynamic topologies, where Long Short-Term Memory (LSTM) networks were integrated into the Actor and Critic components to leverage temporal state information and enhance adaptation to rapid topology changes. The authors in [14] proposed a harmonic-function-based potential field approach to address interference-aware multi-hop routing in dynamic UAV networks. The framework incorporated two specialized fields: an interference field based on expected transmission count and a speed field using auxiliary nodes to handle link instability from UAV movement. An adaptive routing protocol was proposed

in [15] for swarm UAV networks using Mean Field Game theory to enhance link quality and stability. The protocol selected optimal paths based on real-time link conditions and enabled distributed resource management by modeling interactions between individual UAV behaviors and collective swarm dynamics.

B. Low-Latency Multi-Hop Transmission

To deal with the issue of low-latency multi-hop transmission, many researchers have redesigned routing protocols. Reference [16] proposed an energy and throughput management method for a delay-constrained small-world UAV network. It significantly reduced the average path length and number of hops required for data transmission by strategically introducing long-range links between UAVs. The method optimized a cost function considering energy, geographical distance, and hop count. In [17], a UAV-aided multi-access edge computing network with a millimeter-wave backhaul was designed to minimize latency. It first solved an optimal routing problem to determine paths for users, then formulated a joint trajectory and resource allocation problem as a mixed-integer non-convex program. The authors in [18] proposed a layered UAV swarm network architecture for Internet of Things (IoT) services and designed a low-latency routing algorithm. The architecture divided UAVs into upper-layer relay nodes and lower-layer data collection nodes to optimize coverage and connectivity. The algorithm achieved low-latency routing by utilizing partial location information and estimating link stability to dynamically select optimal relay paths with minimal transmission delay. The enhanced Optimized-Link-State-Routing (OLSR)-based scheme proposed in [19] allowed for adaptive link selection in UAV networks. It achieved low-delay routing by calculating optimal paths based on link stability, node energy, bandwidth, and transmission distance.

C. Collaborative Routing in UAV Swarms

To achieve collaborative routing, many schemes dedicated to multi-UAV cooperation have been developed. The authors in [20] proposed a topology-aware routing protocol for UAV swarms to enhance swarm coordination by optimizing inter-UAV routing. It achieved collaborative routing by integrating flight control outputs with network topology awareness, allowing nodes to proactively track formation changes. The protocol dynamically adjusted HELLO intervals and link holding timers based on predicted link duration, calculated from relative UAV positions and movement trajectories. Using bio-inspired algorithms like Ant Colony Optimization and Bee Colony Optimization, the approach in [21] leveraged swarm intelligence to coordinate UAV movements dynamically, enhancing adaptability in highly mobile network topologies to maintain efficient routing paths. Reference [22] implemented an energy-aware collaborative routing scheme for task offloading in UAV swarms. It achieved collaborative routing through a two-stage process: first discovering all possible paths between source and destination UAVs, then selecting optimal paths based on a residual energy maximization score function. Reference [23] proposed an adaptive communication-based

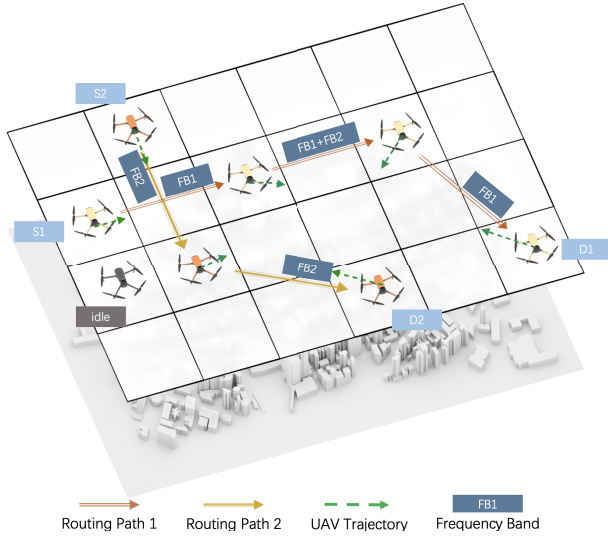


Fig. 1. Concurrent multi-hop transmission in a UAV swarm.

routing algorithm for UAV swarm networks using an MARL approach. It achieved collaborative routing by designing a multi-layer perceptron to determine when routing information flooding was necessary among UAVs, significantly reducing communication overhead.

III. SYSTEM MODEL

As illustrated in Fig. 1, we investigate a UAV swarm network consisting of M UAVs. They are deployed in a gridded airspace, each grid containing at most one UAV. The UAV set is denoted as $\mathcal{M}$, where $|\mathcal{M}| = M$. Consider a transmission period T during which there are K source UAVs $S_1, S_2, \dots, S_K$ requesting to send data to K corresponding destination UAVs $D_1, D_2, \dots, D_K$ in parallel. We denote the k -th source-destination UAV pair as (S_k, D_k) , where $S_k \in \mathcal{M}$ and $D_k \in \mathcal{M}$. The transmission period T is equally divided into N time slots, each lasting for a duration $d_t = \frac{T}{N}$.

Each UAV flies within its own grid at the altitude H and can be treated as quasi-static within a time slot. Without loss of generality, we consider a 3-D Cartesian coordinate system where the location of UAV m at time slot n is written as $\mathbf{q}_m[n] = [\mathbf{x}_m[n], \mathbf{y}_m[n], H]$. The grid containing UAV m is described with four boundary lines $[\mathbf{x}_m^{\min}, \mathbf{x}_m^{\max}, \mathbf{y}_m^{\min}, \mathbf{y}_m^{\max}]$. So we have the following constraints:

$$\mathbf{x}_m^{\min} \leq \mathbf{x}_m[n] \leq \mathbf{x}_m^{\max}, \forall m \in \mathcal{M}, 1 \leq n \leq N, \quad (1)$$

$$\mathbf{y}_m^{\min} \leq \mathbf{y}_m[n] \leq \mathbf{y}_m^{\max}, \forall m \in \mathcal{M}, 1 \leq n \leq N. \quad (2)$$

The initial position of UAV m is given as $\mathbf{q}_m[0] = [\mathbf{x}_m[0], \mathbf{y}_m[0], H]$. The maximum flying speed of a UAV is $V_{\max}$, so its maximum flying distance in one time slot is $S_{\max} = V_{\max} \cdot d_t$, which is formulated as:

$$\|\mathbf{q}_m[n] - \mathbf{q}_m[n-1]\|^2 \leq S_{\max}^2, \forall m \in \mathcal{M}, 1 \leq n \leq N, \quad (3)$$

where $\|\bullet\|$ denotes the Euclidean norm.

We denote the direct communication range of a UAV as κ . Each source-destination pair is assumed to be far enough away that it is incapable of direct communication, so a stable routing path is required to be established between the

source and destination to relay data packages. The routing path keeps transmission within the period. A data package should be relayed by no more than Λ_0 times before it arrives at the destination, and each relay takes one time slot. We use Λ_k to represent how many times the data packages of pair k are relayed, which also means that there are Λ_k direct links between the source-destination pair, so the routing path of pair k can be denoted as a sequence $L_k = [M_k^1, M_k^2, \dots, M_k^{\Lambda_k}, M_k^{\Lambda_k+1}]$, where M_k^i represents the i -th UAV node on the routing path of pair k , M_k^1 the source and $M_k^{\Lambda_k+1}$ the destination, respectively. We introduce the set Ω which includes all possible sequences consisting of elements in $\mathcal{M}$ that is no longer than $\Lambda_0 + 1$, so the following constraints hold for the problem:

$$\Lambda_k \leq \Lambda_0, \forall k \in \{1, \dots, K\}, \quad (4)$$

$$M_k^1 = S_k, M_k^{\Lambda_k+1} = D_k, \forall k \in \{1, \dots, K\}, \quad (5)$$

$$L_k \in \Omega, \forall k \in \{1, \dots, K\}. \quad (6)$$

The UAVs can use U orthogonal frequency bands to transmit data, whose central frequencies and bandwidths are $F_1, F_2, \dots, F_U$ and $B_1, B_2, \dots, B_U$. The transmission power of each UAV is P_0 . When a UAV m is selected as a relaying node, it continuously receives and sends signals in full-duplex mode from the first time slot it receives data packages to the end of the period. We use a binary flag $\Xi_k^\lambda[n]$ to indicate whether the λ -th link of the routing path k has received the first data package and started transmission. $\Xi_k^\lambda[n] = 0, \forall n < \lambda$, which means that the first data package has not yet arrived and the link is not activated, making no interference to others. $\Xi_k^\lambda[n] = 1, \forall n \geq \lambda$, suggesting that the link has started transmission. Otherwise, if a UAV node is not selected as a relay, it lies in the idle state and does not interfere with other links. We introduce a binary variable $\alpha_k^\lambda[u]$ to describe whether the λ -th link of the routing path k transmits data using the frequency band u . $\alpha_k^\lambda[u] = 1$ means that the link adopts frequency band u and $\alpha_k^\lambda[u] = 0$ otherwise. Each link can adopt one or more frequency bands for transmission. Suppose the communication channels between the UAVs are LOS channels following the free-space loss model [24]. Hence, the channel gain from UAV i to UAV j at time slot n is

$$g_j^i[n] = \beta_0 d_j^i[n]^{-2} = \frac{\beta_0}{\|\mathbf{q}_i[n] - \mathbf{q}_j[n]\|^2}, \quad (7)$$

where β_0 means the channel gain value at the reference of $d_0 = 1$ m. $d_j^i[n] = \sqrt{\|\mathbf{q}_i[n] - \mathbf{q}_j[n]\|^2}$ represents the distance between UAV i and UAV j at time slot n . As for the λ -th link in the routing path k , the sending UAV is M_k^λ and the receiving UAV is $M_k^{\lambda+1}$, so the signal-to-interference-plus-noise ratio (SINR) of the λ -th link on frequency band u at time slot n can be formulated as

$$\Gamma_{k,\lambda,u}[n] = \frac{\alpha_k^\lambda[u] \Xi_k^\lambda[n] P_0 g_{M_k^\lambda M_k^{\lambda+1}}^\lambda[n]}{\sum_{\mu=1}^K \sum_{\substack{\nu=1, (\mu,\nu) \notin \{(k,\lambda), (k,\lambda+1)\}}}^{\Lambda_\mu} \alpha_\mu^\nu[u] \Xi_\mu^\nu[n] P_0 g_{M_\mu^\nu M_\mu^{\nu+1}}^\nu[n] + \sigma^2}, \quad (8)$$

where σ^2 means the power of the additive white Gaussian noise (AWGN) at the receiver. The condition $(\mu, \nu) \notin \{(k, \lambda), (k, \lambda + 1)\}$ in the inner summation notation means that the radio waves from M_k^λ and $M_k^{\lambda+1}$ are not included in the interference. The radio waves from M_k^λ are signal to the receiver $M_k^{\lambda+1}$. As for radio waves from $M_k^{\lambda+1}$, we assume that self-interference is eliminated by techniques such as antenna isolation [25]. So the transmission rate of the λ -th link in routing path k at time slot n can be represented as

$$R_k^\lambda[n] = \sum_{u=1}^U B_u \log_2(1 + \Gamma_{k,\lambda,u}[n]). \quad (9)$$

The data packages of the UAV node S_k take Λ_k time slots to reach the UAV node D_k . We denote the data size reaching the destination D_k at time slot n as $C_k[n]$. When $n < \Lambda_k$, $C_k[n] = 0$, since the first data package has not yet reached the destination. When $\Lambda_k \leq n \leq N$, due to the bottleneck effect of multi-hop transmission, $C_k[n]$ is limited by the link with the lowest rate of the routing path. In view of the changes in UAV positions, the transmission rate of each link varies in each time slot. Therefore, we need to trace back the previous transmission rates of the preceding links to determine the lowest transmission rate. The data size reaching the destination D_k at time slot n is expressed as

$$C_k[n] = \min\{d_t \cdot R_k^1[n - \Lambda_k + 1], d_t \cdot R_k^2[n - \Lambda_k + 2], \dots, d_t \cdot R_k^{\Lambda_k}[n]\}, \forall \Lambda_k \leq n \leq N, \quad (10)$$

which can be transformed into the following constraints:

$$C_k[n] \leq d_t \cdot R_k^i[n - \Lambda_k + i], \quad \forall k \in \{1, \dots, K\}, \forall \Lambda_k \leq n \leq N, \forall 1 \leq i \leq \Lambda_k, \quad (11)$$

so the system throughput over the period can be represented as $\sum_{k=1}^K \sum_{n=1}^N C_k[n]$.

In order to improve concurrent multi-hop transmission efficiency, we design a cross-layer concurrent (CLACO) transmission framework that jointly considers network-layer routing paths, link-layer resource allocation, and UAV trajectories. The CLACO framework maximizes system throughput rather than the capacity of a single routing path, which helps the UAV swarm reduce routing path conflicts and decrease co-channel interference. We define $\mathbf{L} = \{L_k, \forall k\}$ storing all routing paths, $\mathbf{A} = \{\alpha_k^\lambda[u], \forall k, \lambda, u\}$ meaning the resource allocation policies and $\mathbf{Q} = \{\mathbf{q}_m[1], \dots, \mathbf{q}_m[N], \forall m \in \mathcal{M}\}$ representing UAV trajectories. The problem is formulated as

$$(P1) : \max_{\mathbf{L}, \mathbf{A}, \mathbf{Q}} \sum_{k=1}^K \sum_{n=1}^N C_k[n], \quad (12a)$$

$$\text{s.t. } \mathbf{x}_m^{\min} \leq \mathbf{x}_m[n] \leq \mathbf{x}_m^{\max}, \forall m \in \mathcal{M}, 1 \leq n \leq N, \quad (12b)$$

$$\mathbf{y}_m^{\min} \leq \mathbf{y}_m[n] \leq \mathbf{y}_m^{\max}, \forall m \in \mathcal{M}, 1 \leq n \leq N, \quad (12c)$$

$$\|\mathbf{q}_m[n] - \mathbf{q}_m[n-1]\|^2 \leq S_{\max}^2, \quad \forall m \in \mathcal{M}, 1 \leq n \leq N, \quad (12d)$$

$$\alpha_k^\lambda[u] \in \{0, 1\}, \forall k \in \{1, \dots, K\}, \quad \forall \lambda \in \{1, \dots, \Lambda_k\}, \forall u \in \{1, \dots, U\}, \quad (12e)$$

$$M_k^1 = S_k, M_k^{\Lambda_k+1} = D_k, \forall k \in \{1, \dots, K\}, \quad (12f)$$

$$L_k \in \Omega, \forall k \in \{1, \dots, K\}, \quad (12g)$$

$$\Lambda_k \leq \Lambda_0, \forall k \in \{1, \dots, K\}, \quad (12h)$$

$$C_k[n] = 0, \forall k \in \{1, \dots, K\}, \forall n < \Lambda_k, \quad (12i)$$

$$C_k[n] \leq d_t \cdot R_k^i[n - \Lambda_k + i], \quad \forall k \in \{1, \dots, K\}, \forall \Lambda_k \leq n \leq N, \forall 1 \leq i \leq \Lambda_k. \quad (12j)$$

This is a joint mixed-integer nonlinear programming problem [26], which is difficult to solve directly. The three sets of variables have distinct characteristics: the set $\mathbf{L}$ consists of unfixed-length sequences of several elements, the set $\mathbf{A}$ includes integer variables, and the set $\mathbf{Q}$ contains continuous variables. These three different sets of variables are interdependent and influence each other. Hence, we decouple the problem into three sub-problems, each sub-problem optimizing one set of variables with the other two fixed.

As shown in Fig. 2, the CLACO framework adopts three modules to separately deal with the three sub-problems. We use block coordinate descent (BCD) [27] to obtain the approximate optimal solution by executing the three modules iteratively. The iteration terminates when the increase of system throughput is smaller than a given threshold $\chi > 0$. However, simply updating the three sets of variables in a straightforward loop may not output an excellent solution. This is because the update direction becomes unstable when all variables change significantly, thus leading to the early stop of the iteration. Taking into account the fact that routing paths determine the potential interference strength, we design a routing-prioritized joint iterative scheme. Firstly, we iteratively update the three sets of variables for a few iterations ψ . Then we output the routing paths in advance and iterate UAV trajectories and resource allocation policies based on them. This framework effectively reduces the instability of update directions and steadily improves the system throughput.

Considering the transmission delay and fault tolerance requirements in real scenarios, we adopt a semi-centralized deployment approach. Some UAVs are selected as local heads based on the network topology, responsible for making decisions on resource allocation, trajectories, and routing schemes for nodes in their neighborhood. The local heads periodically collect and share information on UAV positions and transmission requests within their respective neighborhoods via a control channel. Once the optimization is finalized, each local head broadcasts instructions to the nodes in its neighborhood.

The framework requires initializing UAV trajectories and resource allocation. We set the initial UAV trajectory to remain stationary at the starting point and allow all UAVs to communicate using all frequency bands, i.e.,

$$\mathbf{q}_m^0[n] = \mathbf{q}_m^0[0], \forall m \in \mathcal{M}, 1 \leq n \leq N, \quad (13)$$

$$\alpha_k^\lambda[u]^0 = 1, \forall 1 \leq k \leq K, 1 \leq \lambda \leq \Lambda_k, 1 \leq u \leq U. \quad (14)$$

The routing path module has to deal with combinatorial optimization, which is NP-hard. In large-scale UAV swarm networks, it is impractical to find the optimal solution within a limited time using accurate algorithms like Exhaustive Search

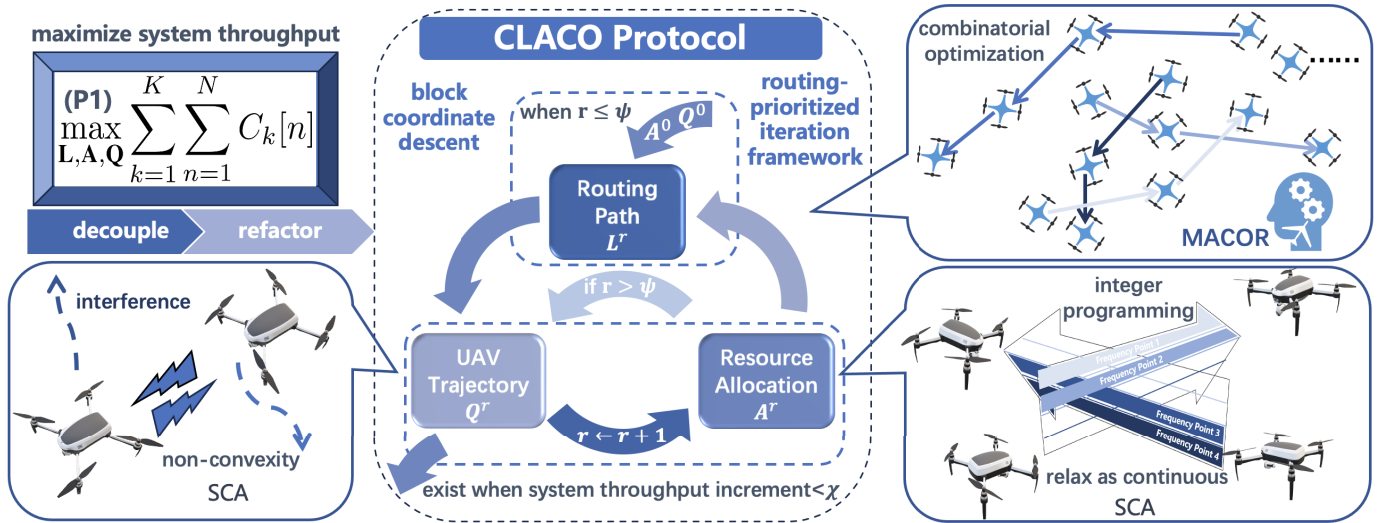


Fig. 2. CLACO adopts a routing-prioritized iterative scheme where the routing paths $\mathbf{L}$ are output in advance and remain stable after a few iterations ψ . Then the resource allocation policies $\mathbf{A}$ and UAV trajectories $\mathbf{Q}$ are optimized in turn based on the routing paths, until the increase of system throughput is smaller than a given threshold $\chi > 0$.

and Dynamic Programming. Given that planning concurrent routes requires coordinating multiple paths, we adopt MARL to promote collaboration. Moreover, by learning the impact of actions on long-term rewards, the MARL approach is well-suited for sequential decision-making involved in multi-hop routing. This module is implemented in Section IV.

The resource allocation module needs to handle integer programming. A general method is to first relax the variables as continuous and then revert them. Next, we deal with the non-convexity using SCA. Similarly, the UAV trajectory module also adopts SCA to address the non-convexity on $\mathbf{Q}$. The two modules are demonstrated in Section V.

IV. MARL-BASED COLLABORATIVE ROUTING

In this section, we propose an MARL-based collaborative routing (MACOR) algorithm to optimize the set of routing path sequences $\mathbf{L}$ with the other two sets of variables $\mathbf{A}$ and $\mathbf{Q}$ fixed. The sub-problem requires finding a combination of routing paths that maximizes the system throughput, where each routing path is an unfixed-length sequence of nodes.

Inspired by the travelling salesman problem [28], we design the MACOR algorithm to work out an approximate optimal solution for routing paths. In the UAV swarm scenario, the first data package from each source node can be viewed as an agent that needs to find a routing path to the destination, and the following data packages from the source node just walk along the existing path. Unlike multi-agent path finding (MAPF) problems [29], in our concurrent routing process, once an agent passes through a certain node, it continues occupying the link and the communication resources until the end of the period, where interference may occur between the links. Moreover, due to wireless transmission, agents can move to more distant nodes in one step, not just adjacent ones. This scenario is a homogeneous environment where all the agents have the same capabilities, target, and operating conditions, so centralized training and decentralized execution (CTDE)

are applied. The training process is conducted on the ground, where the environmental interaction trajectories of all agents are aggregated into a single dataset to train a unified model via parameter sharing. During execution, the model is deployed offline on UAVs to make routing node selections.

For convenience, we use the routing path's index k to refer to the agent on it. In each time slot, if the agent k has arrived at the destination, it stops forwarding and its task is successfully terminated. If not, the agent k selects a neighbouring node and moves to it. If the agent fails to reach the destination in Λ_0 time slots, its task is unsuccessfully terminated. Those agents that have not yet reached the destination nodes or exceeded the maximum forwarding times are called unterminated agents. If two or more agents walk through the same node, they share the communication resources equally.

A. Definition of Markov Decision Process

In each time slot, all the unterminated agents make routing decisions and move forward one step in parallel. The progress can be viewed as a decentralized Partially Observable Markov Decision Process (Dec-POMDP). The Dec-POMDP terminates when all agents are terminated either successfully or unsuccessfully. We define the Dec-POMDP by the tuple $(S, \mathcal{A}_x, \mathcal{P}, \mathcal{R}, \Omega_x, \gamma)$:

- S is the state space, which includes the distribution of all UAV nodes, all the source-destination pairs, all agents' locations, and the existing routing path sequences.
- $\mathcal{A}_x$ is the joint action space, which includes all agents' local action space. In this paper, the action of an agent is to select a neighbouring UAV node as the next hop. Clearly, this action space is discrete. To distinguish neighboring UAV nodes, we refer to them with their grids' relative positions to the agent's current grid, since there is at most one UAV in a grid. Denote the number of grids in a UAV's communication range as Z , so their relative positions can be described by the vectors $\vec{\zeta}_1, \vec{\zeta}_2, \dots, \vec{\zeta}_Z$.

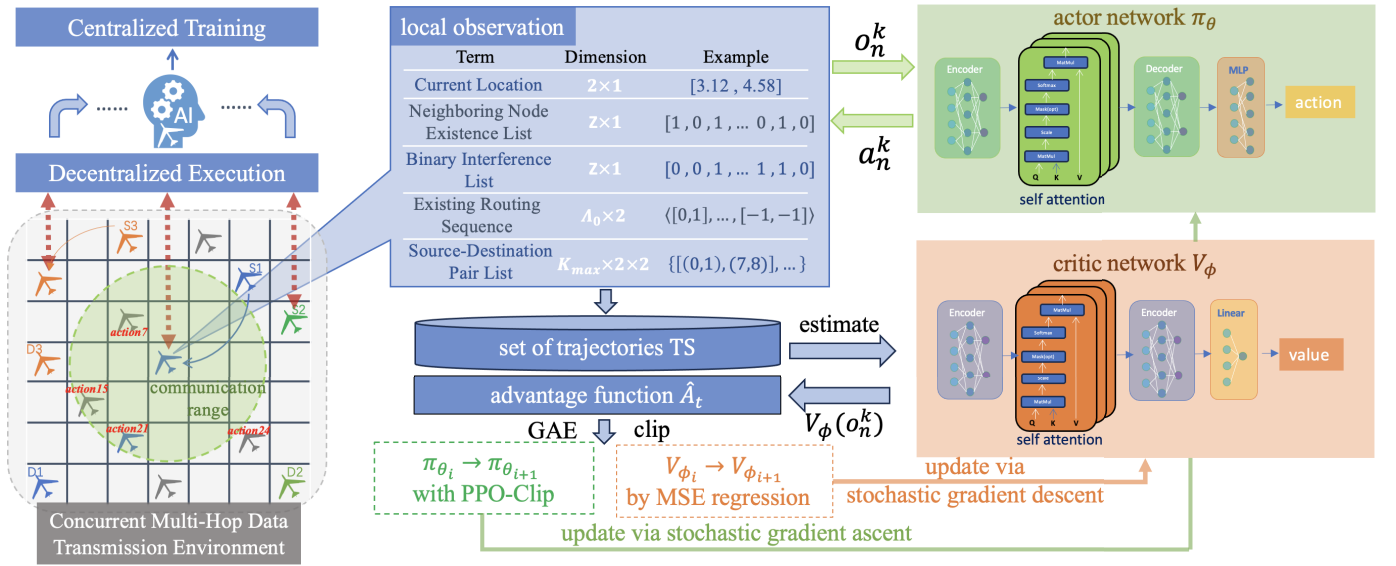


Fig. 3. In the concurrent multi-hop transmission environment, the first data package from each source node is viewed as an agent that looks for a path to the corresponding destination. Considering its distinctions from traditional MAPF problems, we redesign the observation space, action space, and reward function. PPO is applied to update the policy and value networks.

As a result, the action space is $\{\vec{\zeta}_1, \vec{\zeta}_2, \dots, \vec{\zeta}_Z\}$. In other words, we extend the four directions of the traditional path-finding problem to Z . The action performed by agent k at time slot n is a_n^k . When there is no UAV in a neighbouring grid, we disable the corresponding action by using action masks.

- $\mathcal{P}$ is the transition probability function (or transition matrix) modeling the state transition dynamics, which equals 1 since no stochastic process is involved.
- $\mathcal{R}$ is the reward function. To encourage cooperation and avoid local optimality, we need to accurately evaluate the contribution of each agent's action. As a result, we let the agents act asynchronously in each time slot, and design the local reward function of an agent as the increment of the system throughput brought by its action. Specifically speaking, when the agent k performs an action a_n^k at time slot n , we recalculate the system throughput $\sum_{k=1}^K \sum_{n=1}^N C_k[n]$ on the existing paths. The unterminated agents contribute no throughput, but the nodes on their existing paths cause interference. We use $\sum_{i=1}^K \sum_{n=1}^N \hat{C}_k[n]$ on behalf of the old system throughput before a_n^k , so the reward of a_n^k is:

$$r_n^k = \sum_{i=1}^K \sum_{n=1}^N C_k[n] - \sum_{i=1}^K \sum_{n=1}^N \hat{C}_k[n]. \quad (15)$$

- $\Omega_{\times}$ is the joint observation space, which consists of the local observation spaces of all agents. The local observation of the agent k at time slot n contains two categories of data, i.e., period-level data and time-slot-level data, which is demonstrated in Fig. 3. The period-level data is made up of the neighbouring node existence list NNE and the source-destination pair list SDP. The $Z \times 1$ neighbouring node existence list NNE uses 1 to represent that the grid in direction $\vec{\zeta}_z$ contains a selectable UAV node and 0 otherwise. The $K_{max} \times 2 \times 2$

source-destination pair list SDP contains the coordinates of all source-destination pairs, where K_{max} is the maximum number of concurrent routing paths that MACOR can handle. When the number of concurrent routing paths is less than K_{max} , we pad its unused spaces with -1 to preserve the fixed dimensionality of the observation space. The time-slot-level data involves the current location CLO_n^k , the existing routing sequence ERS_n^k , and the binary interference list BIL_n^k of agent k . The 2×1 current location CLO_n^k is the coordinate of the current UAV node. The $\Lambda_0 \times 2$ existing routing sequence ERS_n^k stores in turn the previous node coordinates it passes by. When the agent has not reached the destination or the final path length is less than Λ_0 , we also pad its unused spaces with -1 . The $Z \times 1$ binary interference list BIL_n^k adopts 1 to describe that one of the Z neighbouring nodes interferes with the current node and 0 otherwise. The observation of agent k at time slot n is formulated as:

$$o_n^k = \{NNE, SDP, CLO_n^k, ERS_n^k, BIL_n^k\}. \quad (16)$$

- $\gamma \in [0, 1]$ is the discount factor, which determines the importance of future rewards. With this, the actual return G_n^k after action a_n^k is formulated as

$$G_n^k = \sum_{t=0}^{\Lambda_k-n} \gamma^t r_{n+t}^k, \quad (17)$$

which represents the global accumulated reward after action a_n^k and can be used in value estimation.

B. Policy Optimization Algorithm

In the concurrent routing environment of UAV swarms, the observations and rewards vary significantly across different tasks. To avoid performance degradation caused by excessive policy updates, we utilize PPO [30] to optimize the policy, which is demonstrated in Fig. 3.

PPO optimizes decision-making with a policy network π parameterized by θ and a value network V parameterized by ϕ . It updates the policy network π_θ by maximizing a clipped advantage function. Given the policy parameters θ_i which have been updated for i times, the advantage function $A^{\pi_{\theta_i}}(o_n^k, a_n^k)$ measures the additional benefit of taking the action a_n^k in the state o_n^k compared to average situations, which is shown as:

$$A^{\pi_{\theta_i}}(o_n^k, a_n^k) = \sum_{l=0}^{\Lambda_k - n} (\gamma\lambda)^l \delta_{n+l}^k, \quad (18)$$

where $\delta_t^k = r_t^k + \gamma V_\phi(o_{t+1}^k) - V_\phi(o_t^k)$ is the temporal difference (TD) error at time t and λ is a generalized advantage estimation (GAE) parameter that controls the trade-off between bias and variance. So the optimization objective L^{policy} is

$$L^{\text{policy}}(\theta, o_n^k, a_n^k) = \min \left(\frac{\pi_\theta(a_n^k | o_n^k)}{\pi_{\theta_i}(a_n^k | o_n^k)} A^{\pi_{\theta_i}}(o_n^k, a_n^k), g(\epsilon, A^{\pi_{\theta_i}}(o_n^k, a_n^k)) \right), \quad (19)$$

where

$$g(\epsilon, A) = \begin{cases} (1 + \epsilon)A & A \geq 0 \\ (1 - \epsilon)A & A < 0 \end{cases}, \quad (20)$$

and ϵ is a hyperparameter that controls the range of allowed policy updates. The advantage function is clipped in L^{policy} to prevent large deviations from the current policy. Then the policy parameters θ are updated by taking multiple steps of mini-batch Stochastic Gradient Descent as follows:

$$\theta_{i+1} = \theta_i + \alpha_\theta \nabla_\theta L^{\text{policy}}(\theta_i, o_n^k, a_n^k). \quad (21)$$

The value network V_ϕ is updated to provide accurate value estimates $V_\phi(o_n^k)$ by minimizing the mean squared error (MSE) between the predicted value $V_\phi(o_n^k)$ and the actual return G_n^k , whose loss function is given by:

$$L^{\text{value}}(\phi, o_n^k) = \frac{1}{2} \mathbb{E} \left[(V_\phi(o_n^k) - G_n^k)^2 \right], \quad (22)$$

while the value parameters ϕ are updated by minimizing the loss function $L^{\text{value}}(\phi, o_n^k)$ using gradient descent:

$$\phi_{i+1} = \phi_i - \alpha_\phi \nabla_\phi L^{\text{value}}(\phi_i, o_n^k). \quad (23)$$

In order to elaborate on the MACOR algorithm, we present the details of the training process in Algorithm 1.

V. OPTIMIZING RESOURCE ALLOCATION AND UAV TRAJECTORY WITH SCA

In this section, we respectively solve the resource allocation variables A and the UAV trajectory variables Q adopting SCA.

A. Resource Allocation Optimization

The sub-problem of resource allocation involves integer variables, which can be first relaxed as continuous and then restored to integers by splitting the frequency

Algorithm 1 MARL-Based Collaborative Routing (MACOR)

```

1: INPUT: Initial parameters  $\theta_0, \phi_0$ , learning rates  $\alpha_\theta, \alpha_\phi$ 
2: OUTPUT: Optimized policy parameter  $\theta^*$  continuously
3:  $\theta = \theta_0, \phi = \phi_0$ 
4: for each episode do
5:   Generate UAV distribution and communication pairs
6:   Create new empty set TS
7:   for  $n = 0$  to  $\Lambda_0$  do
8:     for each agent  $k$  do
9:       if unterminated then
10:        Get observation  $o_n^k$ 
11:        Select action  $a_n^k$  using  $\pi_\theta(a_n^k | o_n^k)$  and move
12:        if arrive at destination  $D_k$  or  $n == \Lambda_0$  then
13:          terminate
14:        end if
15:        Get reward  $r_n^k$  and new observation  $o_{n+1}^k$ 
16:        Store trajectory  $(o_n^k, a_n^k, r_n^k, o_{n+1}^k)$  in TS
17:      end if
18:    end for
19:  end for
20:  Update parameters of policy network and value network using trajectories in TS:

```

$$\theta^* \leftarrow \theta + \alpha_\theta \nabla_\theta L^{\text{policy}}(\theta, \mathbf{TS})$$

$$\phi^* \leftarrow \phi - \alpha_\phi \nabla_\phi L^{\text{value}}(\phi, \mathbf{TS})$$

21: **end for**

bands into sub-frequency bands. But the problem is still non-convex due to constraint (12j). So we decide to work out an approximate optimal solution with SCA. Denote $A^r = \{\alpha_k^\lambda[u]^r\}$ as the resource allocation variables of the r -th iteration. For convenience, we use the function $f_{k,\lambda,u,n}(A)$ to represent the transmission rate of the λ -th link in routing path k on frequency band u at time slot n , so $R_k^\lambda[n] = \sum_{u=1}^U f_{k,\lambda,u,n}(A)$, where

$$f_{k,\lambda,u,n}(A) = B_u \cdot$$

$$\log_2 \left(1 + \frac{\alpha_k^\lambda[u] \Xi_k^\lambda[n] P_0 g_{M_k^{\lambda+1}}^\lambda[n]}{\sum_{\mu=1}^K \sum_{\substack{\nu=1, (\mu, \nu) \notin \{(k, \lambda), (k, \lambda+1)\}}}^{\Lambda_\mu} \alpha_\mu^\nu[u] \Xi_\mu^\nu[n] P_0 g_{M_k^{\lambda+1}}^\nu[n] + \sigma^2} \right). \quad (24)$$

Then we use first-order Taylor expansion [31] of $f_{k,\lambda,u,n}(A)$ at A^r as an approximate representation:

$$f_{k,\lambda,u,n}^{\text{apro}}(A^r) = f_{k,\lambda,u,n}(A^r) + \sum_{\mu=1}^K \sum_{\nu=1}^{\Lambda_\mu} \frac{\partial f_{k,\lambda,u,n}(A)}{\partial \alpha_\mu^\nu[u]} \Big|_{A=A^r} (\alpha_\mu^\nu[u] - \alpha_\mu^\nu[u]^r), \quad (25)$$

where

$$\frac{\partial f_{k,\lambda,u,n}(A)}{\partial \alpha_\mu^\nu[u]} \Big|_{A=A^r} = \begin{cases} \frac{B_u \Xi_k^\lambda[n] P_0 g_{M_k^{\lambda+1}}^{M_k^\lambda}[n]}{\ln 2 \left(\sum_{\mu=1}^K \sum_{\nu=1}^{\Lambda_\mu} \alpha_\mu^\nu[u]^r \Xi_\mu^\nu[n] P_0 g_{M_k^{\lambda+1}}^{M_k^\nu}[n] + \sigma^2 \right)} & \text{if } \mu = k \wedge \nu = \lambda, \\ \frac{-B_u \alpha_k^\lambda[u]^r P_0^2 \Xi_k^\lambda[n] \Xi_\mu^\nu[n] g_{M_k^{\lambda+1}}^{M_k^\lambda}[n] g_{M_k^{\lambda+1}}^{M_k^\nu}[n]}{\ln 2 c_{k,\lambda,u,n} \left(c_{k,\lambda,u,n} - \alpha_k^\lambda[u]^r \Xi_k^\lambda[n] P_0 g_{M_k^{\lambda+1}}^{M_k^\lambda}[n] \right)} & \text{if } (\mu, \nu) \notin \{(k, \lambda), (k, \lambda + 1)\}. \end{cases} \quad (26)$$

is the partial derivative of $f_{k,\lambda,u,n}(A)$ with respect to the variable $\alpha_\mu^\nu[u]$ at $A = A^r$ and

$$c_{k,\lambda,u,n} = \sum_{\mu=1}^K \sum_{\nu=1}^{\Lambda_\mu} \alpha_\mu^\nu[u]^r \Xi_\mu^\nu[n] P_0 g_{M_k^{\lambda+1}}^{M_k^\nu}[n] + \sigma^2 \quad (27)$$

is a constant. As a result, we can rewrite constraint (12j) as

$$C_k[n] \leq d_t \cdot \sum_{u=1}^U f_{k,i,u,n-\Lambda_k+i}^{appro}(A^r), \quad \forall k \in \{1, \dots, K\}, \forall \Lambda_k \leq n \leq N, \forall 1 \leq i \leq \Lambda_k. \quad (28)$$

Then we can solve the problem with CVX.

B. UAV Trajectory Optimization

The sub-problem of UAV trajectory is non-convex due to constraint (12j). Denote $Q^r = \{\mathbf{q}_m^r[n], \forall m \in \mathcal{M}, \forall n\}$ to be the UAV trajectory variables of the r -th iteration. Here we use the function $h_{k,\lambda,u,n}(Q)$ to represent the transmission rate of the λ -th link of routing path k on frequency band u at time slot n , so $R_k^\lambda[n] = \sum_{u=1}^U h_{k,\lambda,u,n}(Q)$, where

$$h_{k,\lambda,u,n}(Q) = B_u \cdot \log_2 \left(1 + \frac{\frac{\alpha_k^\lambda[u] \Xi_k^\lambda[n] P_0 \beta_0}{\|\mathbf{q}_{M_k^\lambda}[n] - \mathbf{q}_{M_k^{\lambda+1}}[n]\|^2}}{\sum_{\mu=1}^K \sum_{\substack{\nu=1, (\mu, \nu) \notin \\ \{(k, \lambda), (k, \lambda+1)\}}}^{\Lambda_\mu} \frac{\alpha_\mu^\nu[u] \Xi_\mu^\nu[n] P_0 \beta_0}{\|\mathbf{q}_{M_\mu^\nu}[n] - \mathbf{q}_{M_k^{\lambda+1}}[n]\|^2} + \sigma^2} \right). \quad (29)$$

Next we approximate $h_{k,\lambda,u,n}(Q)$ at Q^r adopting first-order Taylor expansion:

$$\begin{aligned} h_{k,\lambda,u,n}(Q) &\approx h_{k,\lambda,u,n}(Q^r) \\ &+ \sum_{\mu=1}^K \sum_{\nu=1}^{\Lambda_\mu+1} \frac{\partial h_{k,\lambda,u,n}(Q)}{\partial \mathbf{q}_{M_\mu^\nu}[n]} \Big|_{Q=Q^r} (\mathbf{q}_{M_\mu^\nu}[n] - \mathbf{q}_{M_\mu^\nu}^r[n]) \\ &= h_{k,\lambda,u,n}^{appro}(Q^r), \end{aligned} \quad (30)$$

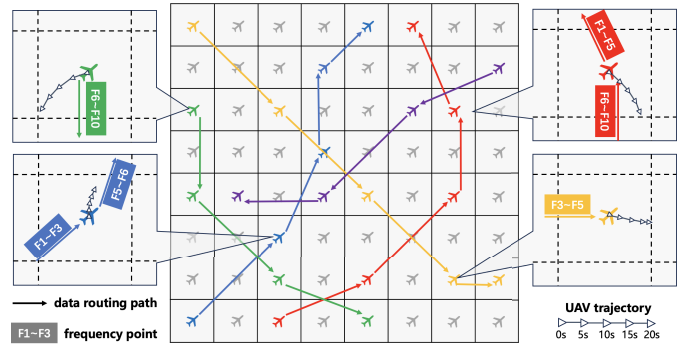


Fig. 4. A representative case with 5 concurrent routing paths.

where

$$\frac{\partial h_{k,\lambda,u,n}(Q)}{\partial \mathbf{q}_{M_\mu^\nu}[n]} \Big|_{Q=Q^r} = \begin{cases} \frac{-2B_u \cdot w_{k,\lambda,u,n}(Q^r)}{\ln 2 \cdot z_{k,\lambda,n}(Q^r) [z_{k,\lambda,n}(Q^r)^2 + w_{k,\lambda,u,n}(Q^r)]} & \text{if } \mu = k \wedge \nu = \lambda, \\ \frac{\iota_{k,\lambda,u,n}(Q^r) - \xi_{k,\lambda,u,n}(Q^r)}{2B_u P_0 \beta_0 \alpha_k^\lambda[u] \Xi_k^\lambda[n] (\mathbf{q}_{M_\mu^\nu}^r[n] - \mathbf{q}_{M_k^{\lambda+1}}^r[n]) \cdot \tau_{k,\lambda,u,n}(Q^r)} & \text{if } \mu = k \wedge \nu = \lambda + 1, \\ \frac{2B_u P_0 \beta_0 \alpha_k^\lambda[u] \Xi_k^\lambda[n] (\mathbf{q}_{M_\mu^\nu}^r[n] - \mathbf{q}_{M_k^{\lambda+1}}^r[n]) \cdot \tau_{k,\lambda,u,n}(Q^r)}{\ln 2 \cdot \rho_{k,\lambda,u,n}(Q^r)^2 \cdot (\rho_{k,\lambda,u,n}(Q^r) + P_0 \beta_0 \alpha_k^\lambda[u] \Xi_k^\lambda[n])} & \text{if } (\mu, \nu) \notin \{(k, \lambda), (k, \lambda + 1)\}. \end{cases} \quad (31)$$

is the partial derivative of $h_{k,\lambda,u,n}(Q)$ with respect to the variable $\mathbf{q}_{M_\mu^\nu}[n]$ at $Q = Q^r$, and $w_{k,\lambda,u,n}(Q^r)$, $z_{k,\lambda,n}(Q^r)$, $\iota_{k,\lambda,u,n}(Q^r)$, $\xi_{k,\lambda,u,n}(Q^r)$, $\rho_{k,\lambda,u,n}(Q^r)$, $\tau_{k,\lambda,u,n}(Q^r)$ are all constants. So constraint (12j) can be rewritten as

$$C_k[n] \leq \sum_{u=1}^U h_{k,i,u,n-\Lambda_k+i}^{appro}(Q^r), \quad \forall k \in \{1, \dots, K\}, \forall \Lambda_k \leq n \leq N, \forall 1 \leq i \leq \Lambda_k. \quad (32)$$

Now the problem is convex and can be solved with CVX.

VI. SIMULATION RESULTS

In order to assess the proposed CLACO framework, in this section, we display its performance within a simulated environment. First, we present the details of the simulation environment and training configuration of the MACOR algorithm. Subsequently, we introduce performance metrics and benchmarks. Finally, we conduct a comprehensive analysis based on multi-dimensional experimental data.

A. Simulation Setup

We consider a 8km×8km airspace which is equally divided into 8×8 grids, each grid containing at most one UAV. In each test, we deploy UAVs of random quantities ranging from 20 to 64 within the airspace, and generate a random number of source-destination pairs ranging from 1 to 10. Then we adopt the CLACO framework to optimize concurrent routing paths, resource allocation, and UAV trajectories for a transmission period. Fig. 4 demonstrates a representative case utilizing

TABLE I
SIMULATION PARAMETERS

Parameter	Value
transmission period T	2 s
number of time slots N	2000
UAV flying altitude H	100 m
UAV maximum fly speed V_{max}	30 m/s
UAV communication range κ	3000 m
maximum routing time Λ_0	10
number of frequency bands U	10
central frequency $F_1, \dots, F_{10}$	2 GHz, ..., 2.9 GHz
bandwidth $B_1, B_2, \dots, B_{10}$	20 MHz
transmission power P_0	35 dBm
reference channel gain β_0	-50 dB
AWGN σ^2	-110 dBm
routing path update time ψ	3
iteration threshold χ	10^{-4}

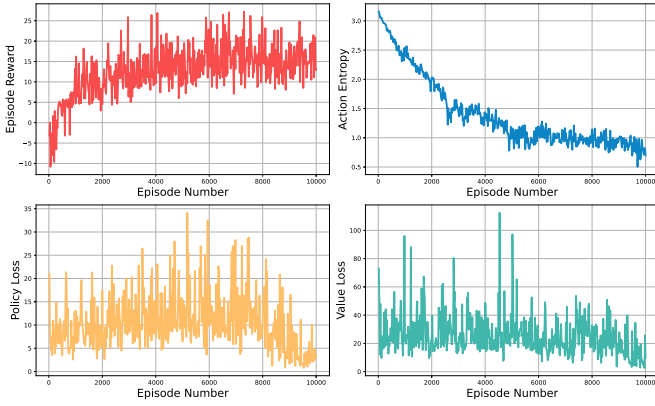


Fig. 5. Convergence process of episode reward, action entropy, value loss, and policy loss.

CLACO, which contains 64 UAVs and 5 concurrent routing paths. The other detailed settings are provided in Table I.

B. Training Details of MACOR

The number of training episodes for MACOR is set as 10000, where each episode involves planning concurrent paths of the UAV swarm network for a period. The training samples include concurrent path numbers ranging from 1 to 10 and UAV node numbers ranging from 20 to 64. Based on the direct communication range κ of 3000m, the action space Z is defined as 24 (i.e., excluding the middle node within a 5×5 field of view). The learning rates of the actor network and critic network $\alpha_\theta, \alpha_\phi$ are set to 1×10^{-5} , respectively. The batch size is 64, the epoch number per update is 20, the discount factor γ is 0.99, the GAE parameter λ is 0.95, and the policy update control hyperparameter ϵ is 0.1. Fig. 5 illustrates the convergence process of different training metrics, including episode reward, action entropy, value loss, and policy loss. Collectively, these training metrics indicate that the model has learned stable and efficient concurrent routing strategies. Notably, the episode reward remains negative in the first 1000 episodes. This is because we introduced a stage-wise reward function in practical engineering to facilitate

curriculum learning [32] for the model, aiming to alleviate initial pathfinding difficulties and sparse rewards caused by the large action space. In the first 1000 training episodes, the reward function is shaped stage-wise as the negative value of the distance (in kilometers) from the current position to the destination minus 0.1. The subtraction of 0.1 at each step encourages the agent to reach the destination with the fewest steps by penalizing unnecessary lingering. The rapid and stable increase in episode reward further validates the effectiveness of this technique.

C. Comparing Frameworks

In the simulation environment, we compare the four multi-hop transmission frameworks:

- **CLACO (Ours):** This is a cross-layer concurrent transmission framework with the MACOR routing algorithm.
- **CIRT:** [33] A cognitive and intelligent routing transmission framework. The framework uses a Q-learning Aided Intelligent Routing (QAIR) algorithm, which dynamically selects routing paths for UAVs by considering achievable rates and residual energy. As for UAV trajectories, CIRT utilizes the Gauss-Markov model by considering speed and direction to ensure smooth trajectory transitions. It allocates resources with spectrum sensing and avoids interference using dynamic spectrum access.
- **HAPOF:** [14] A harmonic-based potential field transmission framework which includes an Interference-Aware Multi-hop Routing (INAMR) algorithm and a UAV trajectory optimization approach based on the gradient of potential field. INAMR employs a virtual potential field constructed using harmonic functions to quantify the effects of interference and relay data based on the function values. By leveraging the field's gradient, HAPOF dynamically guides UAVs to route away from interference.
- **NON-CL:** Non-cross-layer transmission framework. In this benchmark, the UAV trajectories, link layer, and network layer are independently designed. Ad hoc On-Demand Distance Vector Routing (AODV) [34] is selected as the routing algorithm. AODV is a traditional reactive protocol and uses route discovery to find paths and route maintenance to handle link breaks. Besides, the four-color theorem is employed to allocate frequency bands, which means all frequency bands are equally divided into 4 groups, and each grid uses a group different from the closest grids. In addition, the UAV trajectories are designed to be circles whose radius is $\min \{S_{max}, \frac{N}{2\pi} \cdot S_{max}\}$.

D. Performance Metric Definition

We use three metrics to evaluate multi-hop transmission frameworks together with the routing algorithms in them. Here are the definitions of the metrics.

- **System Throughput:** The system throughput means the summation capacity of all routing paths over the transmission period and is formulated as $\sum_{k=1}^K \sum_{n=1}^N C_k[n]$.

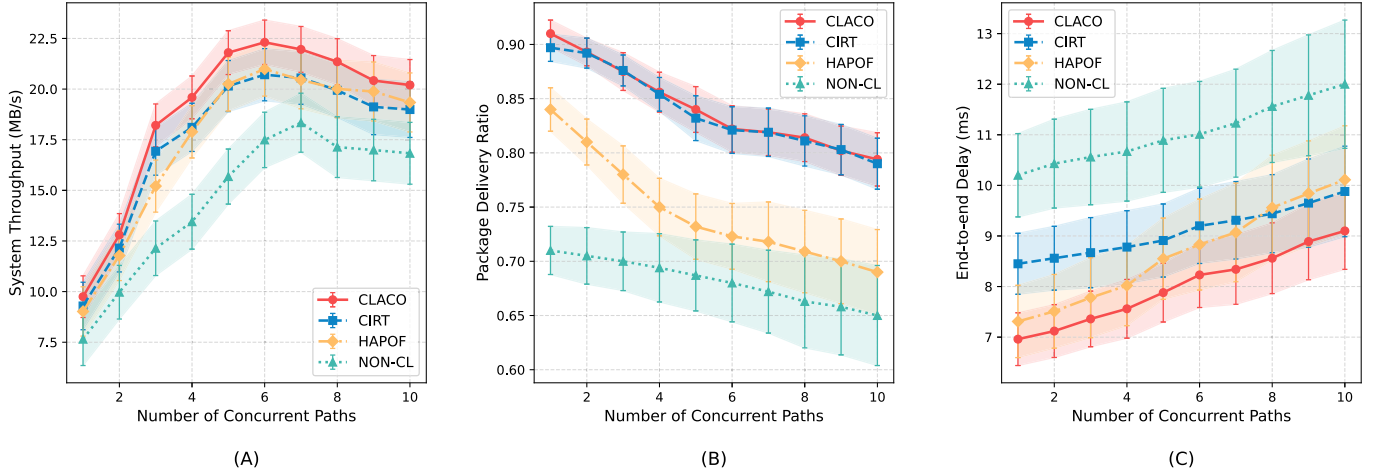


Fig. 6. The system throughput, package delivery ratio, and end-to-end delay of the four multi-hop transmission frameworks as the number of concurrent paths varies.

- **Package Delivery Ratio:** This metric refers to the proportion of the data packages that successfully reach the destination. The package delivery ratio is formulated as:

$$\text{Package Delivery Ratio} = \frac{d_t \cdot \sum_{k=1}^K \sum_{n=1}^N R_k^1[n]}{\sum_{k=1}^K \sum_{n=1}^N C_k[n]}. \quad (33)$$

- **End-to-End Delay:** The end-to-end delay means the average time it takes for a data package to reach the destination. Here, we only take the data packages successfully reaching their destinations into consideration. So the metric can be written as:

$$\text{End-to-End Delay} = d_t \frac{\sum_{k=1}^K \Lambda_k \sum_{n=1}^N C_k[n]}{\sum_{k=1}^K \sum_{n=1}^N C_k[n]}. \quad (34)$$

E. Performance Evaluation

First, we compare the system throughput, package delivery ratio, and end-to-end delay of the four aforementioned frameworks as the number of concurrent paths varies. Then, we compare the performance of the routing algorithms within these frameworks in terms of these three metrics. Finally, we conduct an ablation study to evaluate the contribution of each module in our CLACO framework.

To evaluate the performance of the four frameworks, we generate 100,000 test cases based on the aforementioned simulated UAV network. The number of concurrent paths is varied from 1 to 10, with 10,000 test cases for each path count. Each test case lasts for one period, where all source-destination pairs are randomly generated. We then make the following comparisons based on these test cases.

1) *Concurrent Transmission Framework Evaluation:* Fig. 6 presents the mean value and standard deviation of system throughput, package delivery ratio, and end-to-end delay of the four multi-hop transmission frameworks. The points indicate the mean values, while the error bars display the standard deviations reflecting the distributions of the test results. As for all frameworks, the system throughput firstly goes up

and then slowly declines with the increase of the number of concurrent paths, which is demonstrated in Fig. 6-(A). This is because concurrent transmission can improve system throughput when the network is in low usage, while it aggravates interference between links when the network is congested. Our proposed CLACO framework has an advantage over other transmission frameworks by 7.02%, 7.71%, and 109.26%, which reveals that cross-layer optimization significantly enhances system capacity. In Fig. 6-(B), the package delivery ratio declines with the number of concurrent paths going up due to network congestion. Our proposed CLACO framework and CIRT perform well, where about 85% of data packages are successfully transmitted, while the other two frameworks only reach 75% and 68%. The end-to-end delay is also an important metric of network performance. As shown in Fig. 6-(C), the end-to-end delay of CLACO is about 8 ms, lower than the other benchmarks by 7.60%, 11.94%, and 27.49%. For all metrics, the standard deviations of these four frameworks increase with the number of concurrent paths. This is because a greater number of overlay paths leads to broader variations.

2) *Routing Algorithm Evaluation:* Additionally, we evaluate the routing algorithms in the four transmission frameworks, i.e., MACOR, QAIR, INAMR, and AODV, which have been introduced in Section VI-C. In order to reveal the effect of the routing algorithms, the UAV trajectories and resource allocation strategies are basically set using the initialization methods in Eq. (13) and Eq. (14). The average system throughput, package delivery ratio, and end-to-end delay are shown in Fig. 7, with their standard deviation presented in terms of error bars. The standard deviations of the four routing algorithms across the three metrics do not show significant differences, indicating that the distributions of their experimental data are similar. Compared to other routing algorithms, our MACOR algorithm delivers at least a 19.23% increase in system throughput, a 2.27% increase in packet delivery ratio, and a 5.34% reduction in end-to-end delay.

3) *Ablation Study for CLACO:* Fig. 8 presents the ablation study for the CLACO framework. As for each scheme, the

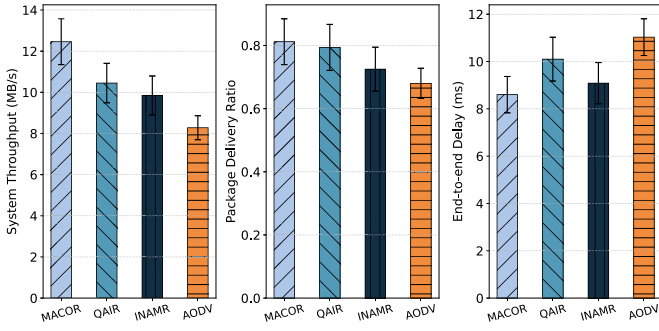


Fig. 7. Routing algorithm evaluation in terms of system throughput, package delivery ratio, and end-to-end delay.

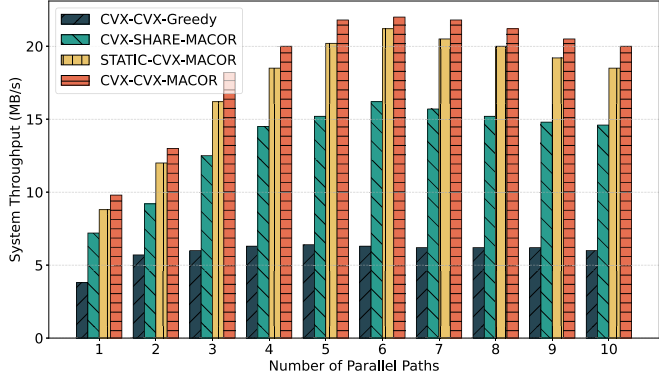


Fig. 8. Ablation study for the CLACO framework. From left to right, each label describes the schemes of UAV trajectory, resource allocation, and routing, respectively.

label from left to right describes the optimization techniques for UAV trajectory, resource allocation, and routing, respectively.

- **CVX-CVX-Greedy:** The MACOR routing algorithm in CLACO is replaced with shortest-path-based routing.
- **CVX-SHARE-MACOR:** The CVX-based resource allocation in CLACO is replaced by allowing UAVs to share all frequency bands.
- **STATIC-CVX-MACOR:** The CVX-based UAV trajectory optimization in CLACO is replaced with keeping all UAVs statically hovering.
- **CVX-CVX-MACOR:** The CLACO framework.

Experimental results indicate that removing the MACOR-based routing module leads to a 68.4% decrease in system throughput, removing the CVX-based resource allocation module leads to a 28.5% decrease, and removing the CVX-based UAV trajectory optimization module leads to a 7.8% decrease. These results demonstrate that the MACOR algorithm plays a critical role in enhancing the performance of the UAV swarm network, while the resource allocation and UAV trajectory optimization modules are also indispensable.

F. Hyperparameter Tuning

In Section III, we propose a routing-prioritized iterative scheme where the routing paths $\mathbf{L}$ are finalized and fixed after ψ iterations to ensure a stable direction for the subsequent updates of resource allocation and UAV trajectories. To validate the advantage of this framework, we compare

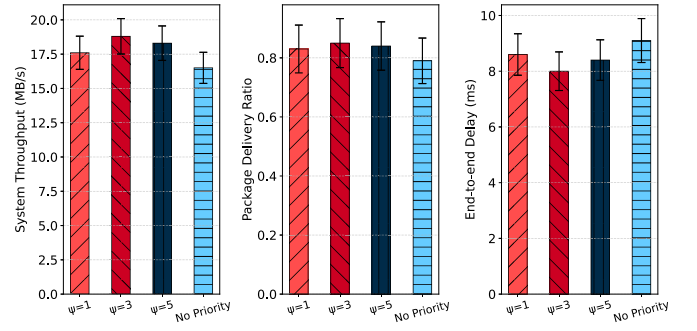


Fig. 9. Performance metrics under different iterative schemes: $\psi = 1$, $\psi = 3$, $\psi = 5$, and No Priority.

the standard iterative scheme with routing-prioritized iterative schemes using different ψ . We compare four schemes: the routing-prioritized approaches with $\psi = 1$, $\psi = 3$, and $\psi = 5$, as well as the standard iterative scheme where routing is not prioritized and all three modules are iterated equally. Fig. 9 shows the system throughput, package delivery ratio, and end-to-end delay of the four schemes, with their mean values presented in the form of bars and standard deviations in error bars. The experimental results demonstrate that the routing-prioritized iterative scheme has significant advantages over the equal iterative scheme in terms of improving system throughput, reducing end-to-end delay, and guaranteeing the packet delivery ratio. The standard deviations of the four schemes do not show significant differences, indicating that the distributions of the experimental data are similar. Specifically, the CLACO framework achieves optimal performance across all metrics when $\psi = 3$.

G. Generalization Analysis

To assess the generalization of the CLACO framework, we examine its network transmission performance with varying UAV deployment ratios and at different swarm network scales.

1) **UAV Deployment Ratio:** The experiments in Section VI-B and VI-F are conducted with 64 UAVs deployed in an airspace of 8×8 grids. Here, the UAV deployment ratio is 100%, which is defined as the ratio of available UAVs to the airspace's total node capacity 64. Nevertheless, there may not always be enough UAV nodes for use. To evaluate the generalization of the frameworks under a declining UAV deployment ratio, we measure the network throughput under 5 concurrent paths with 100,000 test cases by deploying 20, 25, 30, ..., 55, 60, and 64 UAVs, respectively. As shown in Fig. 10, with the decrease of UAV deployment ratio, the system throughput decreases, while CLACO outperforms benchmarks.

2) **UAV Swarm Network Scale:** In addition to the UAV deployment ratio, the scale of the UAV swarm network also affects the performance of concurrent multi-hop transmissions. We measure the end-to-end delay for 5 concurrent transmission paths across different network scales using 100,000 test cases. Here, the network scale is described by the side length of the square airspace, where the side length of each grid is 1 km, the UAV deployment ratio is 100%, and each UAV occupies a single grid. For example, when the airspace side length

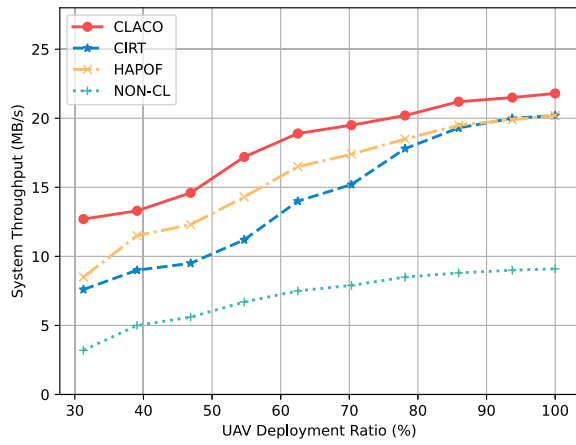


Fig. 10. System throughput under different UAV deployment ratios, which is the ratio of available UAVs to the airspace's total UAV node capacity 64.

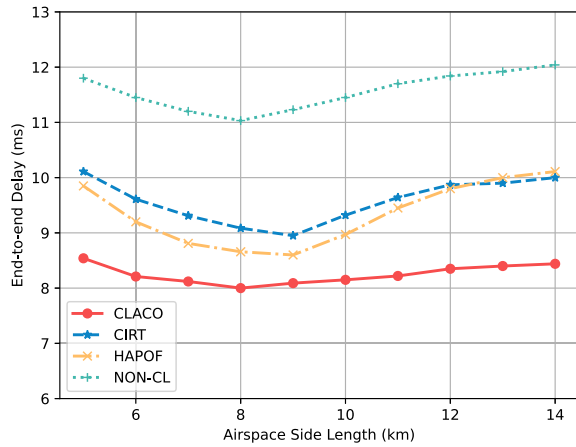


Fig. 11. End-to-end delay under different UAV swarm network scales. The side length of each grid is 1 km, the UAV deployment ratio is 100%, and each UAV occupies a grid.

is 12 km, the UAV swarm network consists of 144 UAVs. As shown in Fig. 11, when the network scale increases, the end-to-end delay first decreases and then increases. This is because, as the network expands, more alternative paths for relay transmission become available, reducing path conflicts and co-channel interference, which initially lowers the delay. However, as the network scale grows further, the destination of a data packet may be farther away, which inevitably leads to an increase in delay. Overall, the CLACO framework consistently maintains the optimal performance.

VII. CONCLUSION

In this paper, to enhance the performance of UAV swarm networks, we propose a cross-layer concurrent multi-hop transmission framework, which maximizes system throughput by jointly optimizing network-layer routing paths, link-layer resource allocation, and UAV trajectories. In view of the complexity of the joint problem, we decouple it into three problems. Then we design three modules and a routing-prioritized joint iterative scheme to solve them in turn. In the routing module, to collectively plan concurrent multi-hop routing paths that require sequential decision-making, we develop an MARL-based collaborative routing algorithm. In the resource allocation module and UAV trajectory module,

we use SCA to deal with non-convexity. Experimental results show that CLACO outperforms other frameworks in system throughput, end-to-end delay, and package delivery ratio.

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Yang Shen (Graduate Student Member, IEEE) received the B.S. degree from Northeastern University, Shenyang, China, in 2022. He is currently pursuing the Ph.D. degree with the School of Computer Science and Technology, Tongji University. His current research interests include UAV communications, wireless resource allocation, and low-altitude transmission protocols.



Bing Li (Member, IEEE) received the Ph.D. degree in computer science and technology from Tongji University, Shanghai, China, in 2021. She is currently an Associate Professor with the School of Computer Science and Technology, Tongji University. She was honored as a "Chengguang Scholar" of Shanghai. She has published more than 40 papers in prestigious journals and conferences and is the author of two academic books. Additionally, she has filed over 20 patents and contributed to the development of three Chinese standards. Her research interests include autonomous UAV swarms, multi-agent reinforcement learning (MARL), and intelligent networking for low-altitude economy.



Rongqing Zhang (Senior Member, IEEE) received the B.S. and Ph.D. degrees (Hons.) from Peking University, Beijing, China, in 2009 and 2014, respectively. He is currently an Associate Professor at The Hong Kong University of Science and Technology (Guangzhou), Guangzhou, China. His research interests include vehicular communications and networking, low-altitude vehicular networks, and connected intelligence. He has authored and co-authored three monographs and over 200 papers in top journals and conferences, with four Best Paper Awards at IEEE ICC 2016, GLOBECOM 2018, ICC 2019, and ICNC 2026. He also received the 2017 First-Class Prize in Natural Science of Ministry of Education of China, the 2023 First-Class Prize in Natural Science of Chinese Association of Automation, and the 2023 First-Class Prize in Natural Science of China Institute of Communications. He is currently the Secretary General of the Connected Intelligence Committee of Chinese Association of Automation, the Vice-Chair of the Information Services Committee of the IEEE ComSoc Asia-Pacific Board, and an Associate Editor of IEEE TRANSACTIONS ON VEHICULAR TECHNOLOGY, IEEE TRANSACTIONS ON NETWORK SCIENCE AND ENGINEERING, and IEEE TRANSACTIONS ON AEROSPACE AND ELECTRONIC SYSTEMS.